

1 Explainability in Reinforcement Learning

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Chapter 1

Clustering Agent Representations

As discussed in the previous chapter, current explainability methods for Reinforcement Learning (RL) still lack intrinsic approaches capable of explaining the internal mechanisms of deep neural policies. To the best of our knowledge, only [Zahavy et al., 2017] have proposed exploiting the Latent Space (LS) learned by neural policies in order to identify the concepts used by the agent. Its main idea is to analyze the Latent Representation (LR) produced by the Neural Network (NN) and organize them into clusters that correspond to high-level concepts.

While the core idea of this paper is closely aligned with our vision of explainability in RL, this approach suffers from two important limitations. First, the clusters are manually constructed. Such a procedure heavily depends on the user’s expertise and neither guarantees reproducibility or scales. Second, no evaluation protocol is proposed to assess the quality of the obtained clusters. Their relevance is essentially determined through visual inspection.

The Clustering Agent Representation (CAR) method proposed in this chapter addresses these two limitations. It automates concept discovery by relying on unsupervised clustering algorithms to identify the Concept Structure (CS) of the LS. It introduces an objective evaluation metric based on the stability of the extracted representations across independently trained policy.

This chapter is organized as follows:

- Section 1.1 lays out the theoretical assumptions underlying the proposed CAR method, which we term the *Representation Convergence Hypothesis* (Hyp. 1).
- Section 1.2 then details the CAR method itself, covering both the extraction of LR using surrogate policies and the automatic detection of CS through clustering.
- Section 1.3 introduces the metrics we propose for evaluation.
- Section 1.4 situates CAR within the broader landscape of mechanistic interpretability.

1.1 Representation Convergence Hypothesis

The CAR method relies on a assumption, referred to as the *Representation Convergence Hypothesis* (Hyp. 1). This hypothesis follows the two assumptions established previously throughout this manuscript.

The *Mechanistic Hypothesis* (Hyp. ??) rejects the notion that deep NN constitute irreducible black boxes. Adopting a mechanistic perspective, we consider that every process results from a sequence of causal computations. Although these computations may be complex, they remain explainable through the analysis of the internal mechanisms.

The *Concept Hypothesis* (Hyp. ??) states that NN achieve generalization by learning concepts. During optimization, the network progressively organizes its LS into representations that capture the information required to solve the task.

Following these two hypotheses, it becomes possible to explain a NN by studying the organization of its LS. This family of approaches is referred to as mechanistic interpretability. Since the CAR method is grounded in these two hypotheses, it falls within this field of research. More specifically, CAR focuses on a specific aspect of mechanistic interpretability known as universality. Universality refers to the consistency with which a concept is recovered across different models trained on the same task [Templeton et al., 2024].

To provide a explanation for this phenomenon, we developed the *Representation Convergence Hypothesis* (Hyp. 1). This hypothesis states that, when NN are trained to solve the same task, a set of concepts should systematically emerge because these concepts provide a effective way of representing the problem. As a result, independently trained models should develop similar concepts in their learned representations.

Hypothesis 1: Representation Convergence Hypothesis

For a given task, independently trained NN tend to develop similar sets of concepts because these concepts provide an efficient representation for solving the task.

The CAR method is designed to investigate the *Representation Convergence Hypothesis* (Hyp. 1) by assessing whether it can consistently recover similar concepts across independently trained models. The hypothesis therefore serves both as the foundation of the method and as the phenomenon it aims to investigate. If CAR consistently recovers the same set of concepts across different training runs, this set is more likely to reflect invariant properties of the task than model-specific artifacts. Conversely, the absence of representation convergence would provide evidence against the hypothesis and would also question the validity of the CAR approach.

It is important to clarify that this hypothesis does not imply convergence at the parameter level. Independently trained models are expected to converge toward different sets of weights. The proposed hypothesis concerns a abstract level of representation. We assume that the conceptual organization induced by their LR converges toward an equivalent CS, even though the numerical implementation of the networks differ.

The remainder of this chapter develops the methodology used to investigate this hypothesis. The *Clustering Universality* metric, introduced in Subsec-

tion 1.3.3, provides a quantitative measure of the convergence observed between the concepts identified in independently trained models.

1.2 Latent Clustering of Surrogate Policies

Following the work of [Zahavy et al., 2017], we analyze the LS through clustering techniques, placing our approach within the broader field of Deep Clustering. This design choice is motivated by our rejection of the *Superposition Hypothesis* (Hyp. ??) and, more generally, of any view that assumes a linearly organized CS, as discussed in Section ??.

The CAR methodology is composed of two main stages. The first stage trains multiple independent NN, called surrogate policies, which implement the same decision policy. The second stage clusters their LR in order to identify the underlying conceptual organization.

This section is organized as follows. In the first subsection, we justify the use of multiple surrogate policies in order to study the stability of representations. In the second subsection, we detail how these surrogate policies are trained so as to best match the original policy. In the third subsection, we describe how their LR are clustered to extract the CS.

1.2.1 Multiple surrogate policies

The CAR method was developed to test the *Representation Convergence Hypothesis* (Hyp. 1). This requires training several NN independently and comparing their LR. Consistent similarities in the resulting CS would provide empirical support for this hypothesis.

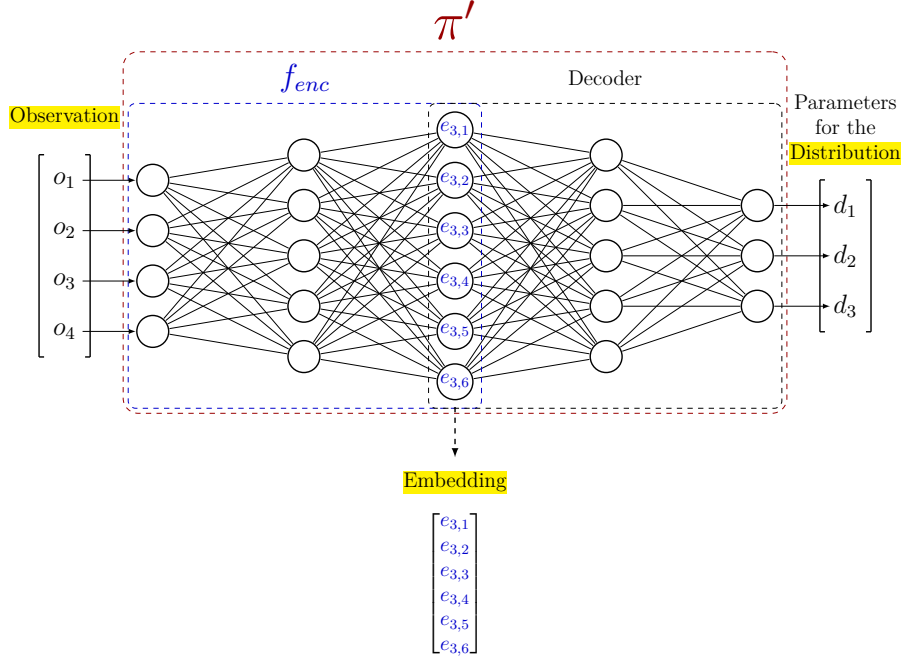
In supervised learning, this protocol is straightforward since several models can be independently trained on the same target function. RL introduces an additional difficulty. Because of stochastic exploration, independently trained agents are not guaranteed to converge toward the same policy. Even if they achieve comparable performance, they may solve the task using different strategies. This makes it difficult to determine whether differences in the LS result from different representations of the same strategy or from differences in the learned strategies themselves.

To overcome this limitation, we consider an already trained initial policy. This policy is then used as a target for imitation learning [Ross et al., 2011]. In this process, several surrogate NN are independently trained in a supervised manner to reproduce its behavior. This makes it possible to investigate whether equivalent policies also exhibit equivalent conceptual organizations.

It is important to emphasize that the use of surrogate policies does not make CAR a surrogate explainability method, as defined in Subsection ??. The surrogate policies remain deep NN. Their role is not to simplify the decision process but to provide multiple independent realizations of the same policy function, enabling the study of representation convergence.

1.2.2 Training surrogate policies

Throughout the remainder of this chapter, the initial policy we aim to explain is denoted as π . This initial policy is obtained without imposing any

Figure 1.1: Example of a surrogate policy architecture π' .

constraints on its origin. It may result from classical or deep reinforcement learning [Sutton and Barto, 2018], alpha-beta algorithm [Knuth and Moore, 1975], or even from demonstrations provided by humans [Ross and Bagnell, 2010]. The crucial requirement is the availability of a dataset containing a sufficient number of projections from the observation space $\mathcal{O}$ to the action space $\mathcal{A}$ induced by π to train a NN.

The replicated policies are referred to as surrogate policies, denoted by π' . For any observation O , the distribution over the action space produced by π' should approximate that of π as closely as possible.

The surrogate policy π' is implemented as a NN parameterized by Θ . This network is composed of two components: an encoder and a decoder, respectively highlighted by the blue and black dashed rectangles in Figure 1.1. The encoder f_{enc} maps the observation O to the embedding. The decoder reconstructs the parameters of the action distribution from this embedding.

To train π' to accurately replicate the action distribution, we define a loss function $\mathcal{L}$ optimized via gradient descent as introduced in Section ???. This loss, defined in Equation (1.1), is computed as the Mean Squared Error (MSE) between the parameters of the action distributions, obtained by applying the function p , which maps a distribution to its parameterization. In practice, this function does not need to be explicitly defined, as neural policies directly output the parameters of the action distribution. Once training is complete, this loss ensures that π' closely approximates the behavior of the original policy π .

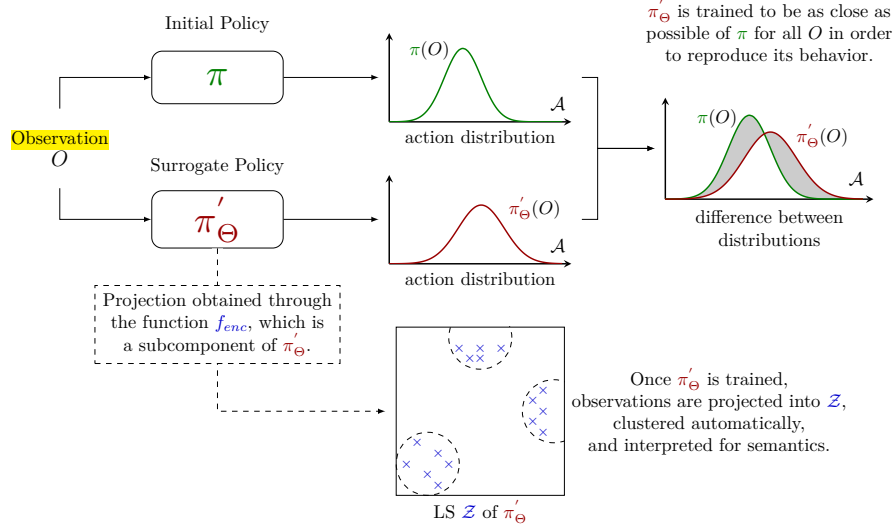


Figure 1.2: Illustration of the Clustering Agent Representation method.

$$\min_{\Theta} \sum_{O \in \mathcal{O}} \mathcal{L}(O) = \min_{\Theta} \sum_{O \in \mathcal{O}} \text{MSE}\left(p(\pi(O)), p(\pi'_{\Theta}(O))\right) \quad (1.1)$$

1.2.3 Cluster extraction

Once the surrogate policies are trained, their encoders are used to generate a point cloud from a set of observations. This point cloud provides a sample of the LS to be analyzed. The proposed CAR method adopts the deep clustering approach introduced in Section ???. Under this approach, conceptually similar observations are expected to produce nearby LR. A CS can therefore be viewed as a region of the LS containing related LR. To identify these regions, a clustering algorithm is applied to the point cloud.

The clustering is performed independently for each of the n surrogate policies. For a surrogate policy π'_i , we denote by $\mathcal{E}_i$ the set of embeddings obtained by applying its encoder f_{enc_i} to a subset of observations $\mathbb{O} \subseteq \mathcal{O}$. A clustering function f_{clust} partitions the point cloud $\mathcal{E}_i$ into groups of nearby LR, assigning each observation a cluster label and yielding the partition $\mathcal{C}_i$. The same observation set $\mathbb{O}$ is used for every policy, so that the partitions $\mathcal{C}_1, \dots, \mathcal{C}_n$ are defined over the same observations and can be compared. This comparison is the basis of the evaluation introduced in the following section. Figure 1.2 and Algorithm 1 summarize the complete CAR procedure.

Two choices govern this analysis without belonging to the clustering algorithm itself. The first is the layer at which the encoder is cut. Any hidden layer defines a LS, and deeper layers produce more abstract LR. CAR does not assume that one layer is correct; it analyzes the LS of whichever layer is selected. The second is the number of clusters. CAR does not assume that a task admits a single correct number of concepts. This number instead sets the granularity at which the conceptual organization is examined. A small number reveals broad, strategic

Algorithm 1: Clustering Agent Representation (CAR)**Input:**Original policy π Observation dataset $\mathbb{O}$ Number of surrogate policies n Surrogate policy architecture π' (encoder f_{enc} , decoder)Learning rate η Clustering function f_{clust} **Output:**Trained surrogate policies $\pi'_{\Theta_1}, \dots, \pi'_{\Theta_n}$ Clusterings $\mathcal{C}_1, \dots, \mathcal{C}_n$ *Behavioral cloning dataset built from the target policy π* **1** $\mathcal{D} \leftarrow \{(O, p(\pi(O))) \mid O \in \mathbb{O}\};$ *MSE loss between action-distribution parameters, as defined in Equation (??)***2** $\mathcal{L} \leftarrow \text{MSE};$ *Independently train n surrogate policies by behavioral cloning***3** **for** $i = 1, \dots, n$ **do****4** Initialize Θ_i randomly;*Behavioral cloning of π via SGD, see Algorithm ??***5** $\Theta_i \leftarrow \text{SGD}(\mathcal{D}, \pi', \mathcal{L}, \eta, \Theta_i);$ *Cluster the latent representations of each surrogate policy***6** **for** $i = 1, \dots, n$ **do***Embeddings produced by the encoder of π'_{Θ_i}* **7** $\mathcal{E}_i \leftarrow \{f_{enc_i}(O) \mid O \in \mathbb{O}\};$ *Partition of the embedded observations into clusters***8** $\mathcal{C}_i \leftarrow f_{clust}(\mathcal{E}_i);$ **9** **Return** $(\pi'_{\Theta_1}, \dots, \pi'_{\Theta_n}), (\mathcal{C}_1, \dots, \mathcal{C}_n)$

167 concepts, while a larger number reveals finer, tactical ones. The influence of
 168 both choices is studied in Chapter ??.

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 172 correct representation. Instead, it analyzes the LS corresponding to the selected
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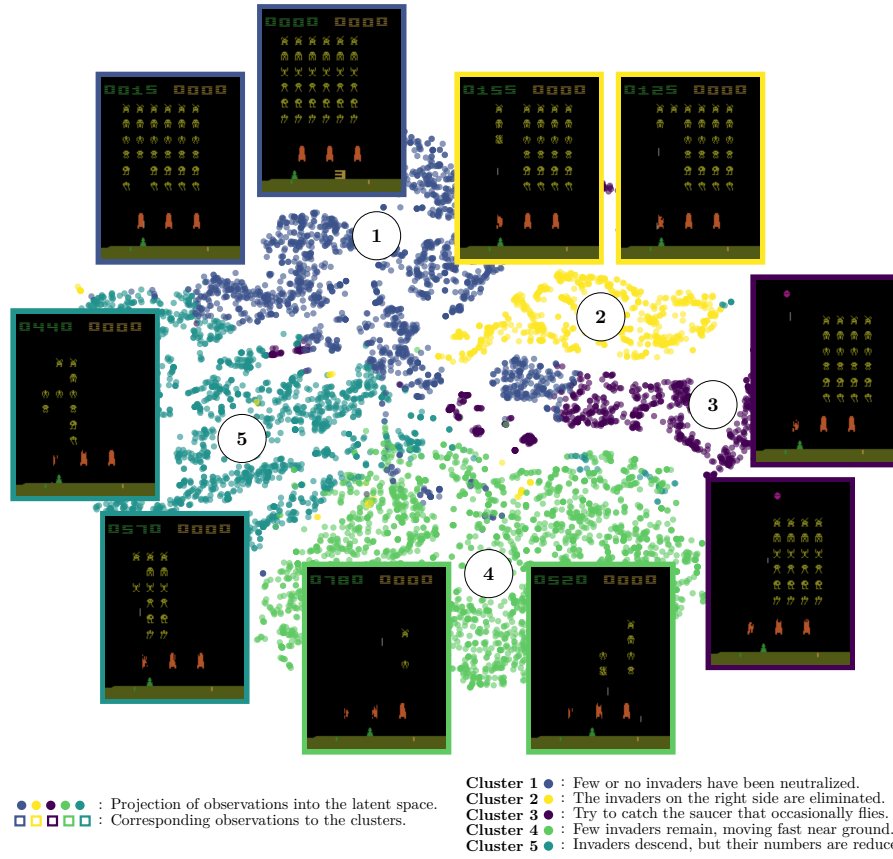


Figure 1.3: Two-dimensional projection of the LS embeddings of a surrogate policy trained on Space Invaders, together with the CAR clusters and their corresponding observations.

Figure 1.3 shows the outcome of this procedure for a surrogate policy trained on Space Invaders, using five clusters. The point cloud is a two-dimensional projection used only for visualization. The clusters are computed in the original high-dimensional LS. Each cluster gathers observations that correspond to a distinct, interpretable stage of a game:

- **Cluster 1:** The start of a game, with few or no invaders neutralized.
- **Cluster 2:** The right-hand group of invaders eliminated.
- **Cluster 3:** The pursuit of the bonus saucer.
- **Cluster 4:** The end of a game, with a few fast invaders close to the ground.
- **Cluster 5:** The middle of a game, with the invaders descending but already thinned out.

The position of an observation in the LS thus reflects the stage of the game it belongs to, suggesting a correspondence between the geometry of the LS and the

semantic properties of game states. To assess this correspondence objectively, we introduce a set of evaluation metrics in the following section.

1.3 Evaluation Metrics

A key contribution of this work is the development of three evaluation metrics, designed to assess the effectiveness of CAR:

- **Behavior Reconstruction** : it measures how well a surrogate policy faithfully reproduces the behavior of the original policy by comparing their cumulative rewards.
- **Clustering Universality** : it evaluates the stability of clusterings by measuring the similarity of clusters obtained from independently trained surrogate policies.
- **Abstraction Score** : it quantifies the extent to which LS clusters capture abstractions rather than directly reflecting the structure of observations or actions.

The remainder of this section details the motivation for each metric and describes how they are calculated. In the first subsection, we develop the *Behavior Reconstruction*. We then introduce the Adjusted Rand Index (ARI) measure, which serves as the foundation for the two other metrics introduced in the subsequent subsections, namely *Clustering Universality* and *Abstraction Score*.

The following notations will be adopted for the rest of the manuscript: we consider n surrogate policies, denoted as $\pi'_1, \dots, \pi'_n$, trained as described in Section 1.2. Each surrogate policy is associated with a projection function $f_{enc_1}, \dots, f_{enc_n}$, which is applied to the set of observations $\mathbb{O}$, yielding embeddings $\mathcal{E}_1, \dots, \mathcal{E}_n$. A clustering function is then applied to each embedding set, producing partitions $\mathcal{C}_1, \dots, \mathcal{C}_n$.

1.3.1 Behavior Reconstruction

We argue that explaining a function b that closely approximates another function a is essentially the same as explaining a . If b accurately captures the key decision patterns of a , understanding b provides meaningful insights into how a works [Guidotti et al., 2018]. In our setting, this means that generating explanations for π' that faithfully reproduce π amounts to explaining π directly. To verify that π' is indeed a faithful copy, we introduce the *Behavior Reconstruction* metric.

This metric is computed by comparing the cumulative rewards obtained by the original policy π and the surrogate policies π' over m evaluation episodes. The behavior reconstruction metric is then defined as:

$$\text{Behavior Reconstruction} = \frac{1}{n \sum_{i=1}^m g_{\pi}^i} \sum_{j=1}^n \sum_{k=1}^m g_{\pi'_j}^k \quad (1.2)$$

In imitation learning, it is unlikely to expect the surrogate policy π' to outperform the initial policy π it aims to approximate through supervised

learning [Ross and Bagnell, 2010]. Consequently this metric ranges in $[0, 1]$. Values approaching 1 indicate minimal loss in reproducing π , while values close to 0 indicate poor reproduction.

We deliberately adopt a reward-based definition rather than alternatives based on actions or action distributions (e.g., training loss, accuracy, KL divergence [Shlens, 2014], or optimal transport [Peyré and Cuturi, 2020]). Since surrogate policies are trained in a supervised manner, they cannot develop strategies beyond those demonstrated by the initial policy. This means that if the rewards achieved are comparable, the surrogate policy is effectively copying the original policy’s behavior.

1.3.2 Adjusted Rand Index

Within CAR, concepts are assumed to correspond to clusters in the LS. Combined with the *Representation Convergence Hypothesis* (Hyp. 1), this leads to a direct experimental prediction: if independently trained surrogate policies converge toward the same conceptual organization, then clustering their LR should produce similar partitions.

When a particular observation is processed by the network, it is expected to activate the same concept regardless of the specific neural implementation of the policy. If independently trained policies converge toward the same conceptual organization, a given observation should activate the same concept across all models. As a consequence, observations grouped together within a cluster for one policy should also be grouped together for another independently trained policy.

In the clustering literature, the most used measures for comparing two partitions is the ARI [Hubert and Arabie, 1985]. This measure quantifies the agreement between cluster assignments independently of the spatial organization of the clusters. Both metrics introduced later in this chapter, *Clustering Universality* and *Abstraction Score*, rely on the ARI. Since the ARI is a complex measure, we review its definition and discuss its main properties.

The ARI is derived from the Rand Index (RI) [Rand, 1971], which measures the agreement between two partitions by considering every possible pair of point cloud. For each pair, four situations are possible: the point may belong to the same cluster in both partitions, to different clusters in both partitions, or they may disagree by being grouped together in only one of the two partitions. The RI measures the proportion of pairs for which the two partitions agree:

$$\text{RI} = \frac{\text{number of agreeing pairs}}{\text{total number of pairs}}.$$

Although the RI is intuitive, it suffers from an important limitation. In most clustering problems, the majority of pairs of points belong to different clusters. Consequently, even two completely independent partitions will agree on a large number of pairs, simply because both assign them to different clusters. This effect becomes more pronounced as the number of points or clusters increases.

The ARI addresses this issue by accounting for the agreement that would be expected purely by chance. It compares the observed agreement RI to the expected agreement under random partitions $\mathbb{E}[\text{RI}]$. The score is then normalized by the maximum agreement achievable beyond this random baseline, $\max(\text{RI}) - \mathbb{E}[\text{RI}]$. As a result, an ARI of 0 corresponds to the level of agreement

274 expected from random clusterings, while an ARI of 1 indicates perfect agreement
275 between the two partitions:

$$\text{ARI} = \frac{\text{RI} - \mathbb{E}[\text{RI}]}{\max(\text{RI}) - \mathbb{E}[\text{RI}]}$$

276 In practice, the ARI is computed directly from the cluster assignments of the
277 two partitions. Let clustering A be composed of partitions $\{A_i\}$ and clustering B
278 of partitions $\{B_j\}$. Let $a_i = |A_i|$ denote the number of samples in partition i of
279 clustering A , and $b_j = |B_j|$ the number of samples in partition j of clustering B .
280 Furthermore, let $n_{ij} = |A_i \cap B_j|$ be the number of samples shared by partition i
281 of clustering A and partition j of clustering B , and let n be the total number of
282 samples. With these definitions, the ARI can be computed as:

$$\text{ARI} = \frac{\sum_{i,j} \binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} \right] / \binom{n}{2}}{\frac{1}{2} \left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2} \right] - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} \right] / \binom{n}{2}}$$

283 Because it is bounded by 1, a value such as 0.5 can be misleadingly interpreted
284 as “halfway to a perfect match.” This interpretation is incorrect because the ARI
285 is normalized against the agreement expected from two random partitions. As a
286 result, values close to 1 require the two clusterings to be nearly identical, while
287 a value of 0.5 already indicates substantial agreement [Amezquita et al., 2020].

288 1.3.3 Clustering Universality

289 Having introduced the ARI, we can now define the *Clustering Universality* metric.
290 While the ARI quantifies the similarity between two conceptual partitions, the
291 objective of CAR is to assess the consistency of conceptual organizations across
292 an entire set of independently trained surrogate policies.

293 To this end, the same set of observations is projected into the LS of each
294 surrogate policy and clustered. Under the *Representation Convergence Hypothesis*
295 (Hyp. 1), these independently obtained partitions are expected to be similar.
296 *Clustering Universality* quantifies this by computing the mean pairwise ARI:

$$\text{Clustering Universality} = \frac{1}{\binom{n}{2}} \sum_{i=1}^n \sum_{j=0}^{i-1} \text{ARI}(\mathcal{C}_i, \mathcal{C}_j). \quad (1.3)$$

297 This metric takes values in $[-1, 1]$. A value close to 1 indicates that surrogate
298 policies produce highly similar partitions, suggesting that the clusters reflect
299 intrinsic properties of the policy. Conversely, a value near 0 implies that the
300 similarity between partitions is no better than that of two randomly generated
301 clusterings, meaning that the clustering fails to capture any meaningful structure
302 in the policy. A negative value, on the other hand, signals that the agreement
303 between partitions is worse than what would be expected by chance, indicating
304 a systematic disagreement between the clusterings.

305 Figure 1.4 illustrates *Clustering Universality* in the ideal case. It shows three
306 LS obtained by projecting the same seven observations, labeled 1 to 7, through
307 three independently trained surrogate policies. Although the embeddings occupy
308 different positions, the clusters group exactly the same elements in all three
309 cases. The partitioning is therefore identical, resulting in an ARI of 1.

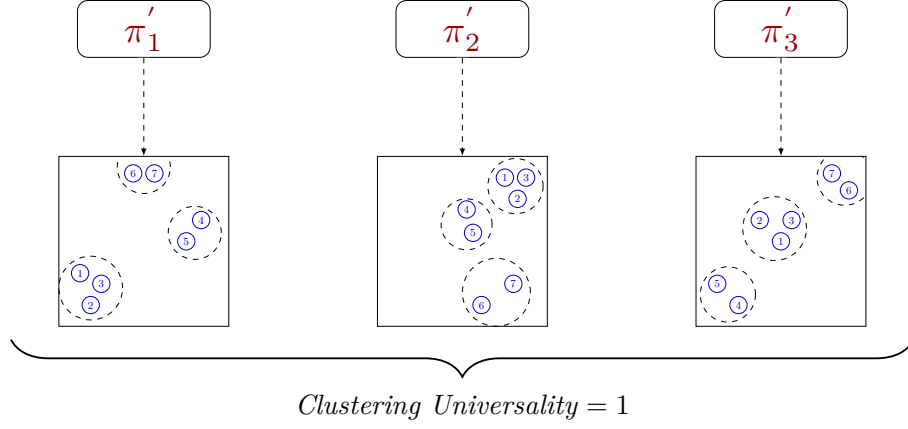


Figure 1.4: Illustration of *Clustering Universality* in the ideal case where projections of the same observations are grouped within the same clusters across all surrogate policies.

1.3.4 Abstraction Score

If *Clustering Universality* is high, it becomes important to determine what underlies this stability. Two possibilities arise:

- (i) the LS merely reproduces the structure of the observation space or action distribution, offering no abstraction, as defined in Definition ??;
- (ii) the LS captures similarities in decision-making processes, yielding behaviorally meaningful clusters.

To distinguish among these cases, we introduce the *Abstraction Score*. This metric contextualizes latent clusterings by examining how their patterns align with reference structures derived from the raw observation and action data. We consider : $\mathcal{C}_{obs}$, the clustering in the observation space $\mathbb{O}$, and $\mathcal{C}_{act}$, the clustering in the action-space distribution parameters. The clusterings $\mathcal{C}_{obs}$ and $\mathcal{C}_{act}$ are performed using the same algorithm, distance metric, and hyperparameters as those used for $\mathcal{C}_i$ in the LS, ensuring that any biases introduced by the clustering procedure itself are minimized.

For each latent clustering $\mathcal{C}_i$, we define its similarity to these baselines using the ARI:

$$A_i^{obs} = \text{ARI}(\mathcal{C}_i, \mathcal{C}_{obs})$$

$$A_i^{act} = \text{ARI}(\mathcal{C}_i, \mathcal{C}_{act})$$

The *Abstraction Score* is subsequently calculated as one minus the average of the maximum absolute similarity values across these two reference baselines:

$$Abstraction\ Score = 1 - \frac{1}{n} \sum_{i=1}^n \max(|A_i^{obs}|, |A_i^{act}|) \quad (1.4)$$

The *Abstraction Score* ranges from $[0, 1]$. A score around 0 means that the latent clusters are strongly aligned with either the observation or the action

space, reflecting minimal abstraction, cases (i). A score close to 1 indicates that the latent clusters are independent of both the observation and action spaces, capturing abstractions that meaningfully reflect the agent’s behavior, case (ii).

The results of applying CAR and these metrics to concrete environments are presented in Chapter ??.

1.4 Conclusion

The CAR method is the concrete instantiation of the three hypotheses developed in this manuscript: the *Mechanistic Hypothesis* (Hyp. ??), the *Concept Hypothesis* (Hyp. ??), and the *Representation Convergence Hypothesis* (Hyp. 1). It assumes that neural policies can be explained through their internal mechanisms, that these mechanisms are organized as concepts in the LS, and that independently trained policies converge toward an equivalent conceptual organization. The *Clustering Universality* and *Abstraction Score* metrics turn these assumptions into measurable quantities, which makes the hypotheses empirically testable. The experimental study reported in Chapter ?? provides support for this hypothesis-driven approach.

Beyond its role in testing these hypotheses, CAR provides a general and automated method for analysing agent representations. Both the analysis procedure and the metrics are fully automated, requiring no manual intervention or environment-specific adjustments. This makes CAR applicable to a wide range of RL settings.

A more fundamental limitation concerns the semantic interpretation of the clusters. While CAR provides a quantitative description of the structure present in the agent’s representations, the meaning assigned to individual clusters still relies on manual interpretation. CAR therefore characterizes how the LS is organized, but not what each concept stands for. Developing methods to automatically extract the semantics of each cluster is an important direction for future work.

When situating CAR within the field of explainability, substantial work remains to be done. The structure it recovers can be considered an explanation in the sense defined in Section ??, since it provides information about the concepts represented in the agent’s LS. How such explanations can be used to support validation, diagnosis, or certification, as discussed in Chapter ??, remains to be established. Connecting CAR explanations more directly to the agent’s decisions and failures would help determine how they can serve these broader objectives.

Acronyms

ARI Adjusted Rand Index 8–11

CAR Clustering Agent Representation 1–3, 5–10, 12

CS Concept Structure 1–3, 5

LR Latent Representation 1–3, 5, 9

LS Latent Space 1–3, 5–12

MSE Mean Squared Error 4, 6

NN Neural Network 1–4

RI Rand Index 9

RL Reinforcement Learning 1, 3, 12

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